🔥 Complete Roadmap: PyTorch & TensorFlow (Beginner → Advanced)

⏱️ মোট সময়: 3–4 মাস

👉 প্রতিদিন 2–3 ঘণ্টা ধরেই প্ল্যান করা

🧱 Phase 0: Prerequisites (1–2 সপ্তাহ)

👉 এটা skip করলে পরে কষ্ট হবে

1️⃣ Python (Must)

- Variables, loops, functions
- List, tuple, dict
- OOP basics (class, object)
- File handling

📌 Practice:

```
for i in range(5):
    print(i*i)
```

2️⃣ Math for Deep Learning

(ভয় পাওয়ার কিছু নাই, applied way)

- Linear Algebra
  - Vector, Matrix
  - Dot product
- Calculus
  - Gradient (concept)
- Probability
  - Mean, Variance
  - Normal Distribution

📌 Real idea:

> Gradient = model কীভাবে ভুল কমাবে তার দিক

3️⃣ NumPy + Pandas

- NumPy arrays
- Broadcasting
- Pandas DataFrame
- Data cleaning

🧠 Phase 1: Deep Learning Core Concepts (1 সপ্তাহ)

👉 Framework ধরার আগে মাথা clear করো

- What is Neural Network?
- Perceptron
- Activation Function
  - ReLU, Sigmoid, Softmax
- Loss Function
  - MSE, Cross-Entropy
- Optimizer
  - SGD, Adam
- Backpropagation (conceptual)

📌 Interview question ready:

> Why ReLU better than sigmoid?

🔥 Phase 2: PyTorch (Beginner → Intermediate) (4–5 সপ্তাহ)

🟢 Week 1: PyTorch Basics

- torch.Tensor
- tensor vs numpy
- GPU (CUDA)
- requires_grad

```
import torch
x = torch.tensor([2.0], requires_grad=True)
y = x**2
y.backward()
print(x.grad)
```

🟢 Week 2: Build Neural Network

- nn.Module
- Linear Layer
- Forward pass
- Loss & Optimizer

```
import torch.nn as nn

class Model(nn.Module):
    def __init__(self):
        super().__init__()
        self.fc = nn.Linear(10, 1)

    def forward(self, x):
        return self.fc(x)
```

🟢 Week 3: Training Loop (Important 🔥)

- Epoch
- Batch
- Zero grad
- backward()
- optimizer.step()

📌 বুঝতে হবে:

> training loop = heart of DL

🟢 Week 4: CNN (Computer Vision)

- Convolution
- Pooling
- Image classification
- Dataset & DataLoader

📌 Project:
✅ Handwritten Digit Recognition (MNIST)

🟢 Week 5: Advanced PyTorch

- Transfer Learning
- Fine-tuning
- Saving & loading model
- Custom Dataset

📌 Project:
✅ Face Mask Detection
✅ Image Classifier (Cats vs Dogs)

⚡ Phase 3: TensorFlow + Keras (3–4 সপ্তাহ)

🟣 Week 1: TensorFlow Basics

- tf.Tensor
- eager execution
- GPU usage

🟣 Week 2: Keras API

- Sequential Model
- Functional API
- Compile, Fit, Evaluate

```
model = tf.keras.Sequential([
    tf.keras.layers.Dense(64, activation='relu'),
    tf.keras.layers.Dense(1)
])
```

🟣 Week 3: CNN with TensorFlow

- ImageDataGenerator
- Data augmentation
- Callbacks

📌 Project:
✅ Image Classifier
✅ Flower Classification

🟣 Week 4: Advanced TensorFlow

- Custom training loop
- TensorBoard
- Model optimization
- tf.data pipeline

🤖 Phase 4: Specialized Deep Learning (Optional but Powerful)

🔹 NLP

- Tokenization
- Word Embeddings
- RNN, LSTM
- Transformers (BERT concept)

📌 Project:
✅ Sentiment Analysis
✅ Fake News Detection

🔹 Time Series

- LSTM
- Forecasting

📌 Project:
✅ Stock Price Prediction

🚀 Phase 5: Deployment & Real World (2 সপ্তাহ)

🔧 Model Deployment

- Save model (.pth / .h5)
- Flask / FastAPI
- Docker (তুমি আগেই জিজ্ঞেস করেছো ✔)

📌 Project:
✅ ML Model API
✅ DL model on cloud

🏆 Phase 6: Portfolio & Career

- GitHub projects
- Kaggle notebooks
- README writing
- Model explanation

🔥 PyTorch vs TensorFlow (When to Use)

PyTorch	TensorFlow
Research	Production
Flexible	Scalable
Debug easy	Deployment strong

👉 **Industry tip:**
✔ Research → PyTorch
✔ Production → TensorFlow

🎯 Final Advice (Bondhu Talk 💙)

- একসাথে দুইটা শিখো না
  👉 আগে **PyTorch**, পরে **TensorFlow**
- Copy-paste না, নিজে লিখো
- Project না করলে শেখা incomplete